GARV ARORA

Robotics & Embedded Systems Engineer | Autonomous Systems | ADAS

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SUMMARY

R&D Lead overseeing 2 autonomous systems teams (50+ engineers) at SEDS VIT, the apex body of SEDS India. Final-year CS undergraduate (IoT) at VIT with 2+ years building ROS2 navigation stacks, CUDA-accelerated perception pipelines, mmWave sensor systems, and ARM bare-metal firmware. Placed 12th (IRC 2025) and 16th (IRC 2026) globally among 100+ international teams; targeting roles in robotics, embedded systems, ADAS, or autonomous vehicle engineering.

EDUCATION

Vellore Institute of Technology | Vellore, India — **CGPA: 8.37 / 10.0**
Bachelor of Technology – Computer Science & Engineering (Specialization: IoT) — *August 2023 – July 2027*

Gems International School | Gurugram, India — **Class 10: 89.3% | Class 12: 76.4%**
Central Board of Secondary Education (CBSE) — *November 2016 – April 2023*

TECHNICAL SKILLS

Languages: C, C++, Python, Java, JavaScript, SQL, Bash, Verilog, C#, HTML/CSS
Robotics & Autonomous Systems: ROS2, Nav2, MoveIt2, RTAB-Map, Micro-ROS, SLAM, EKF/UKF, PID Control, Point Cloud Processing, mmWave Radar
Perception & ML: OpenCV, YOLOv8, MediaPipe, Intel RealSense SDK, PyTorch, TensorFlow, scikit-learn, CUDA, HuggingFace Transformers
Embedded & Infrastructure: ARM Cortex-M, ESP32, I2C/SPI/UART, FreeRTOS, Docker, Git, GitLab CI/CD, Linux, AWS, PyQt, MongoDB

WORK EXPERIENCE

SEDS Projects VIT | SEDS India — **January 2026 – Present**
Research & Development Lead (Team Ardra – Drone | Team Vyadh – Rover) — *Vellore, India*
- Direct R&D roadmap for 2 autonomous systems teams (50+ engineers) — Team Ardra (UAV/drone) and Team Vyadh (planetary rover) — under SEDS India, the national apex body for space engineering students.
- Guided 3 sub-team leads through navigation architecture reviews and embedded sensor debugging, resolving 10+ critical pre-competition issues across Team Ardra (ISDC) and Team Vyadh (IRC/IRDC).

Team Vyadh – Martian Rover | SEDS VIT — **April 2024 – Present**
Autonomous Systems Developer — *Vellore, India*
- Architected a full ROS2 navigation stack (Nav2, RTAB-Map Visual SLAM, RealSense D455) generating real-time 3D elevation maps, cutting manual teleoperation time by 70% in competition.
- Slashed CV pipeline latency by ~60% and raised throughput 4x via CUDA, YOLOv8, ArUco/AprilTag detection, and Linux V4L2 tuning across 4 simultaneous camera streams.
- Flashed Micro-ROS firmware to ESP32 microcontrollers for distributed sensor acquisition; fused IMU and encoder data with an EKF, cutting odometry drift by ~40% and enabling sub-10 cm autonomous positioning.
- Shipped a PyQt5 autonomous mission dashboard with real-time telemetry visualization, parameter tuning, and FSM-based mission sequencing for autonomous competition runs.

Wissen Baum Engineering Solutions — **May 2025 – July 2025**
Software Automation Intern — *Pune, India*
- Authored a BDD test suite (Python, Behave, pytest) covering 40+ scenarios, raising regression coverage from 0% to 100% on 3 core product flows in 6 weeks.
- Accelerated GitLab CI/CD by parallelizing 3 stages (Playwright + Cypress), trimming run time ~35% (18 min to 12 min); replaced manual QA with a pandas/NumPy script processing 10,000+ rows per run.

PROJECTS

mmWave Earthquake Survivor Detection Robot | *ROS2, Python, Hi-Link LD2410C, ESP32* — 2025
- Interfaced a Hi-Link LD2410C mmWave radar with ROS2 to detect survivor breathing signatures through 30 cm of rubble; fused radar point-cloud with IMU odometry for autonomous debris navigation, flagging coordinates with <15 cm error.

Gesture-Controlled 5-DOF Robotic Arm | *C++, Arduino, MPU6050, Flex Sensors* — 2024
- Mapped real-time MPU6050 orientation and 2 flex sensor readings to 5-DOF servo commands with <50 ms latency; trained gesture classifier on 200+ samples, achieving 94% accuracy across 8 gestures.

HayaiOS – Bare-Metal RTOS | *C, ARM Assembly, ARM Cortex-M4* — 2026
- Implemented preemptive scheduler (1 ms tick), context switching in <50 assembly instructions, mutex/semaphore primitives, and a full HAL isolating all register-level I/O on ARM Cortex-M4.

3D Reconstruction – IMU-Enhanced KinectFusion | *C++, OpenCV, Intel RealSense SDK* — 2025
- Stabilized KinectFusion TSDF reconstruction by fusing RealSense D455 depth and IMU via complementary filter, reducing tracking failures by ~50% across a 6-DOF fast-motion test trajectory.

CERTIFICATIONS & ACHIEVEMENTS

IRC 2025: 12th globally (100+ teams) | **IRC 2026:** 16th globally | **IRDC 2025:** Top 5 Finalist | **Oracle Cloud 2025:** Generative AI Professional | Data Science Professional